Deep Convolutional Neural Networks

Feature Extraction vs Fine-tuning with ResNet18

Datasets: CIFAR-10 & STL10

MSc Artificial Intelligence — Deep Learning & Multimedia Data Analysis

1. Introduction

This report investigates transfer learning with deep convolutional neural networks (CNNs), specifically comparing two strategies for adapting a pretrained ResNet18 model to new classification tasks: Feature Extraction and Fine-tuning.

Feature Extraction (FE): All layers of the pretrained ResNet18 are frozen. Only a new fully connected (FC) head (512 → 10) is trained. This results in only 5,130 trainable parameters.

Fine-tuning (FT): The last two residual blocks (layer 3, layer 4) are unfrozen alongside the new FC head, allowing the network to adapt its higher-level representations to the target domain. This results in approximately 10.5M trainable parameters.

Both strategies are evaluated on two benchmark datasets — CIFAR-10 and STL10 — which share the same 10 semantic categories but differ significantly in image resolution and training set size. This cross-dataset comparison allows us to assess how dataset characteristics influence the effectiveness of each transfer learning strategy.

The source code and experimental results are available at the following links:

- ➔ GitHub Repository
- ➔ CIFAR-10 Experiment
- ➔ STL10 Experiment

2. Methodology

2.1 Model Architecture

ResNet18 was used, a well-known deep CNN pretrained on ImageNet. The model includes residual (skip) connections, which help prevent the vanishing gradient problem by adding the input directly to the output of each block: output = F(x) + x.

For both experiments, the original 1000-class classifier was replaced with a new head consisting of a Dropout layer (p=0.3) followed by a linear layer that outputs 10 classes. In the fine-tuning experiments, layers 3 and 4 were additionally unfrozen, allowing the model to adapt its higher-level representations to the target domain.

2.2 Datasets

CIFAR-10 contains 60,000 color images of 32×32 pixels across 10 classes. The training set (50,000 images) is split into 45,000 training and 5,000 validation images. STL10 contains 13,000 labeled images at 96×96 resolution across the same 10 semantic categories, split into 4,500 training, 500 validation, and 8,000 test images.

All images are resized to 224×224 to match the input dimensions expected by ResNet18, and normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Training data is augmented with random horizontal flips and random crops.

2.3 Training Setup

Both models were trained for 15 epochs using the Adam optimizer and Cross-Entropy loss. For Feature Extraction, a learning rate of 1e-3 was used. For Fine-tuning, the learning rate was reduced to 1e-4 to avoid disrupting the pretrained weights. A StepLR scheduler was also applied, halving the learning rate every 5 epochs. The model checkpoint with the highest validation accuracy was saved and used for final evaluation.

3. Results

3.1 Quantitative Comparison

Table 1 summarizes the test accuracy, trainable parameters, and training time for all four experimental conditions.

	FE — CIFAR-10	FT — CIFAR-10	FE — STL10	FT — STL10
Test Accuracy	79.19%	**94.95%**	93.16%	**95.54%**
Accuracy Gain (FT vs FE)	—	**+15.76%**	—	**+2.37%**
Trainable Params (FE)	5,130	—	5,130	—
Trainable Params (FT)	—	10,498,570	—	10,498,570
Training Time	25.9 min	27.5 min	3.2 min	3.4 min
Training Images	45,000	45,000	4,500	4,500
Image Resolution	32×32	32×32	96×96	96×96

Table 1. Summary of results across datasets and training strategies. FE = Feature Extraction, FT = Fine-tuning.

3.2 Learning Curves

The learning curves below show training and validation accuracy per epoch for both datasets.

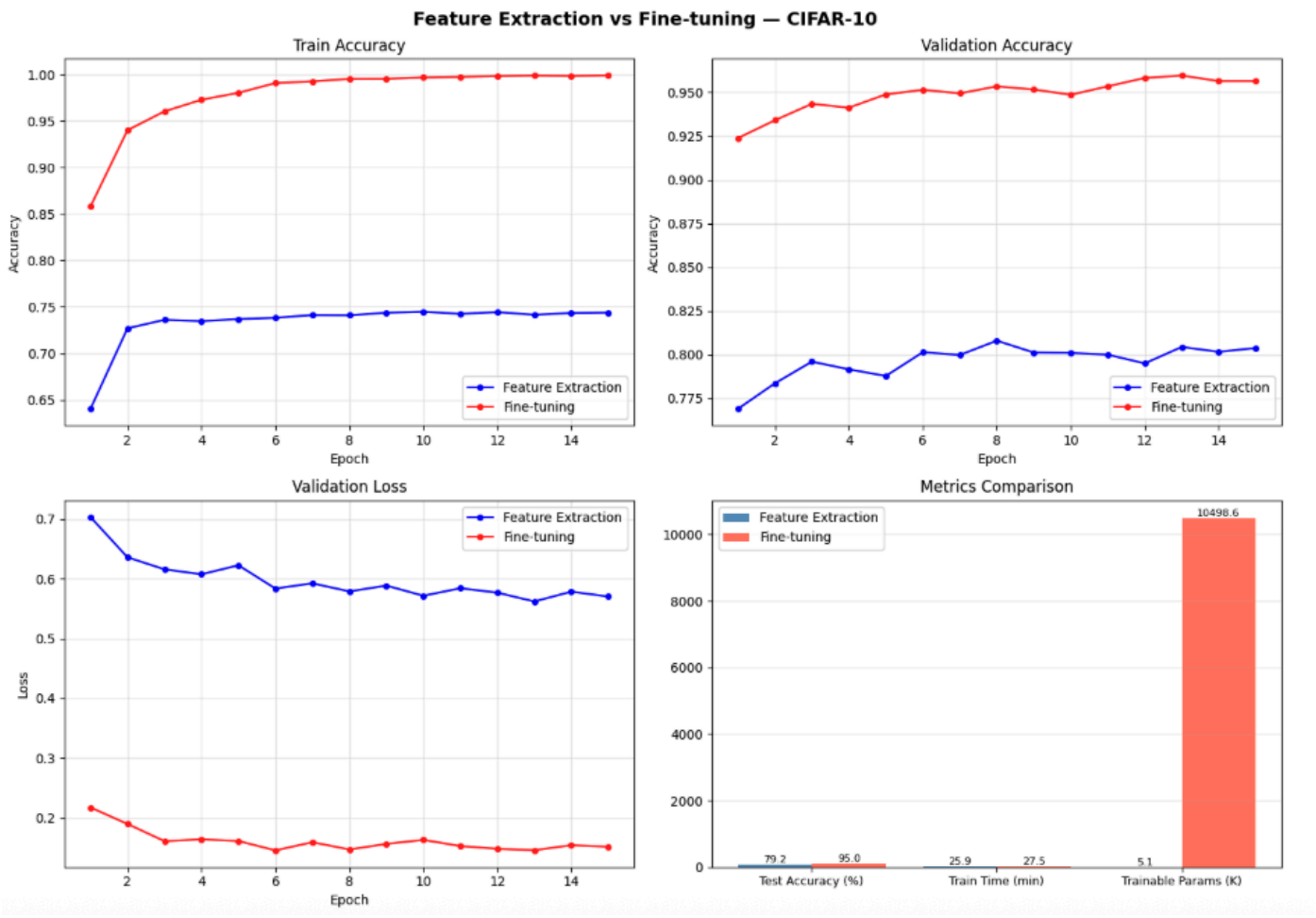


Figure 1. Train/Val accuracy and loss curves for CIFAR-10. Fine-tuning converges significantly faster and achieves higher accuracy.

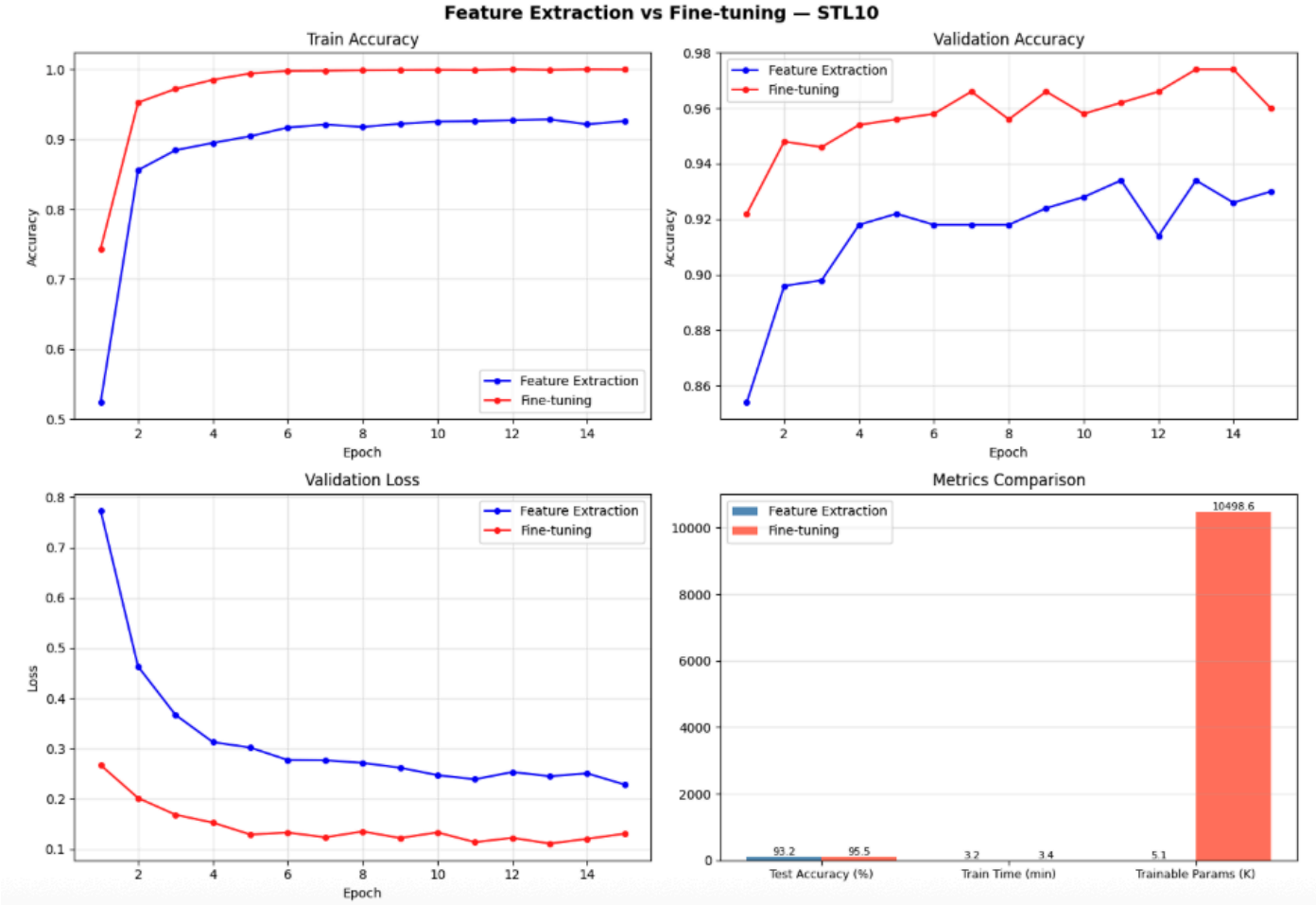


Figure 2. Train/Val accuracy and loss curves for STL10. Feature Extraction performs competitively, narrowing the gap with Fine-tuning.

3.3 Confusion Matrices

The confusion matrices below show per-class classification performance. The most common confusions in both datasets are cat↔dog and deer↔horse, which are visually similar classes.

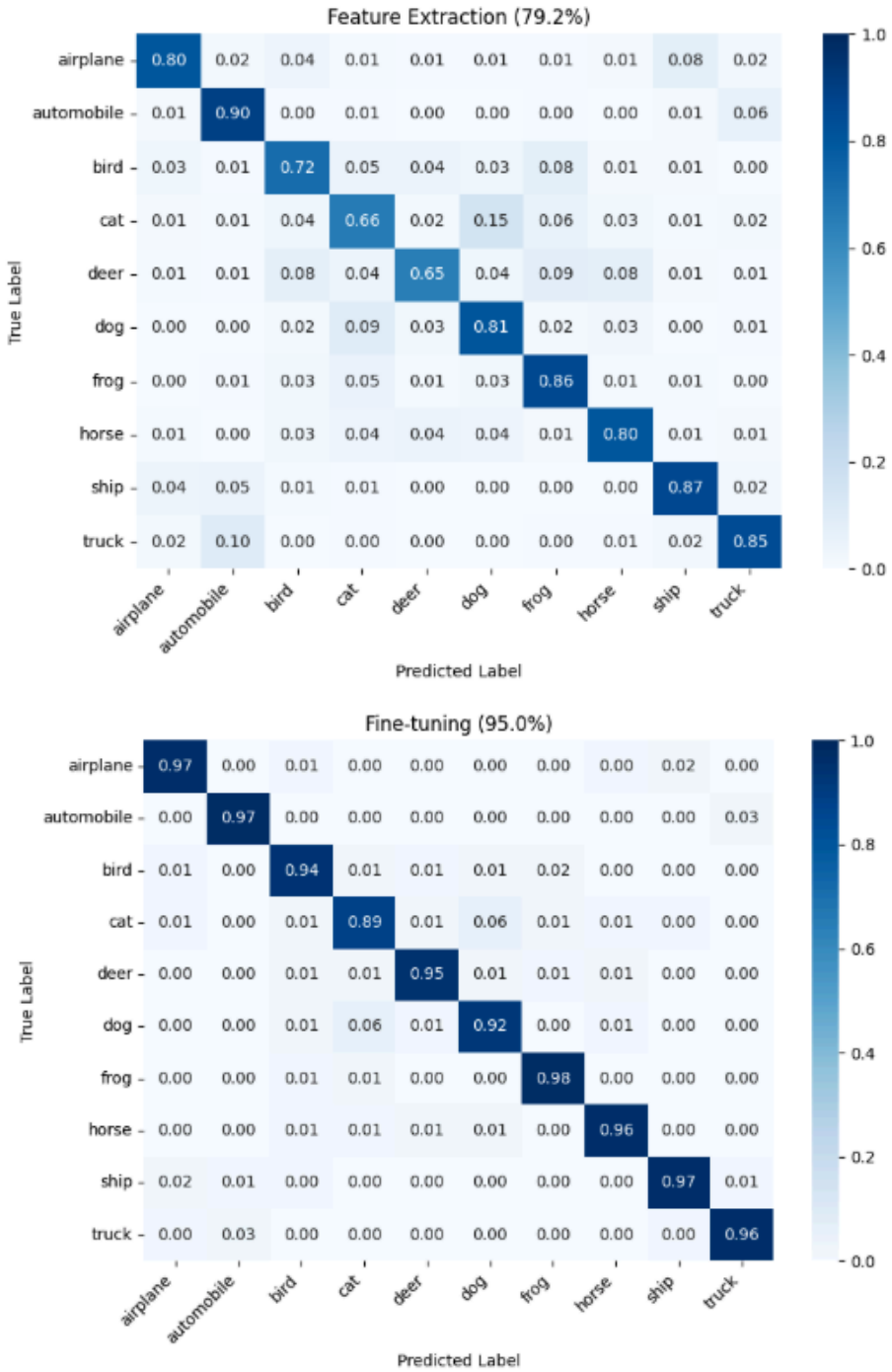
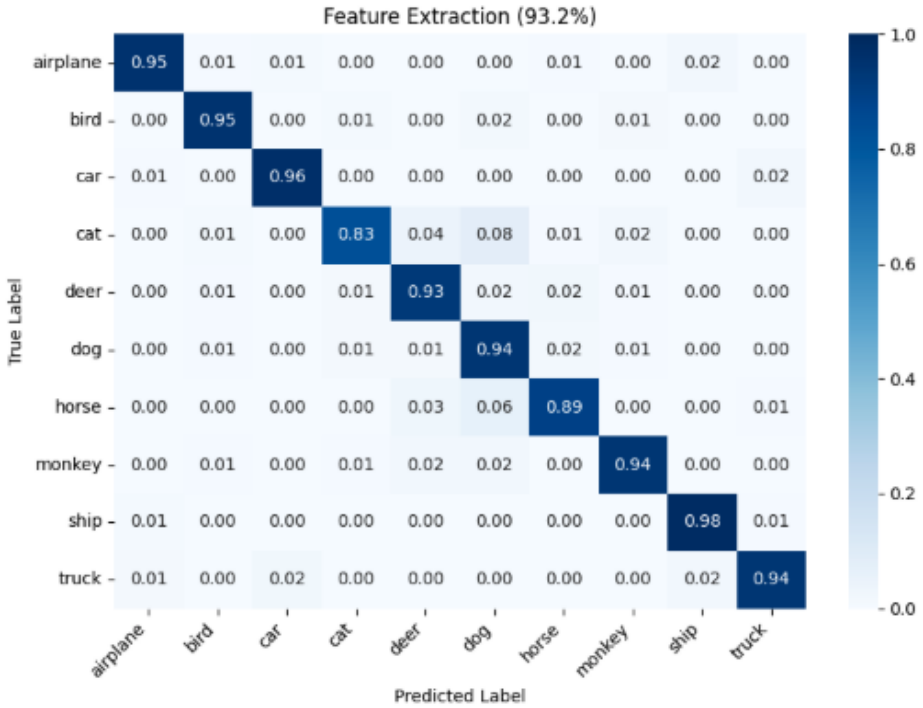


Figure 3. Normalized confusion matrices for CIFAR-10. Fine-tuning reduces misclassifications across all classes.



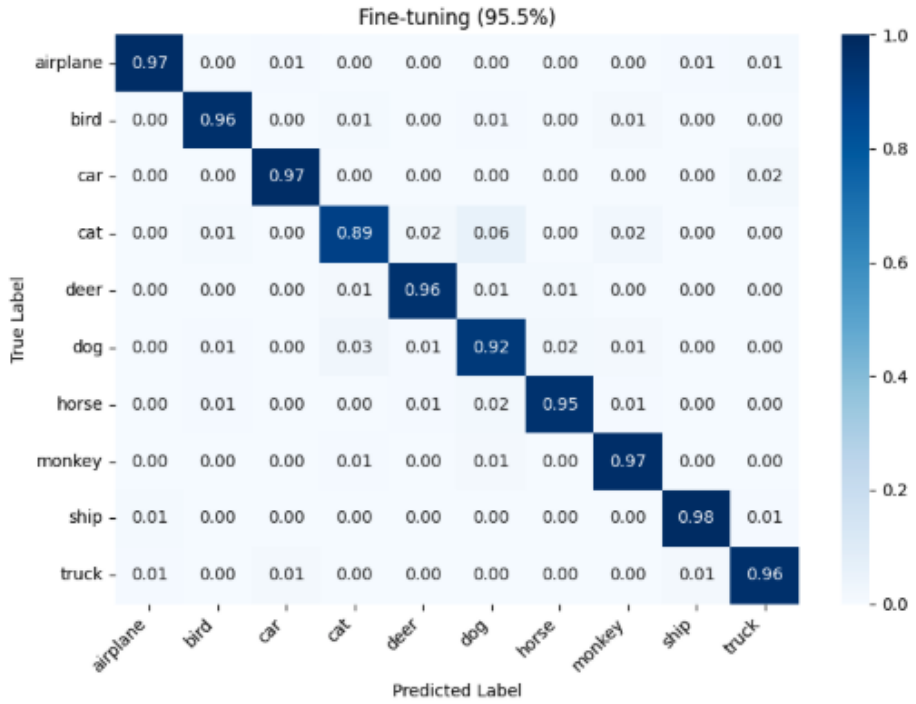


Figure 4. Normalized confusion matrices for STL10. Both models perform well; differences are subtle.

3.4 Grad-CAM Visualizations

Grad-CAM highlights the image regions each model attends to when making a prediction. Warmer colors (red/yellow) indicate higher attention.

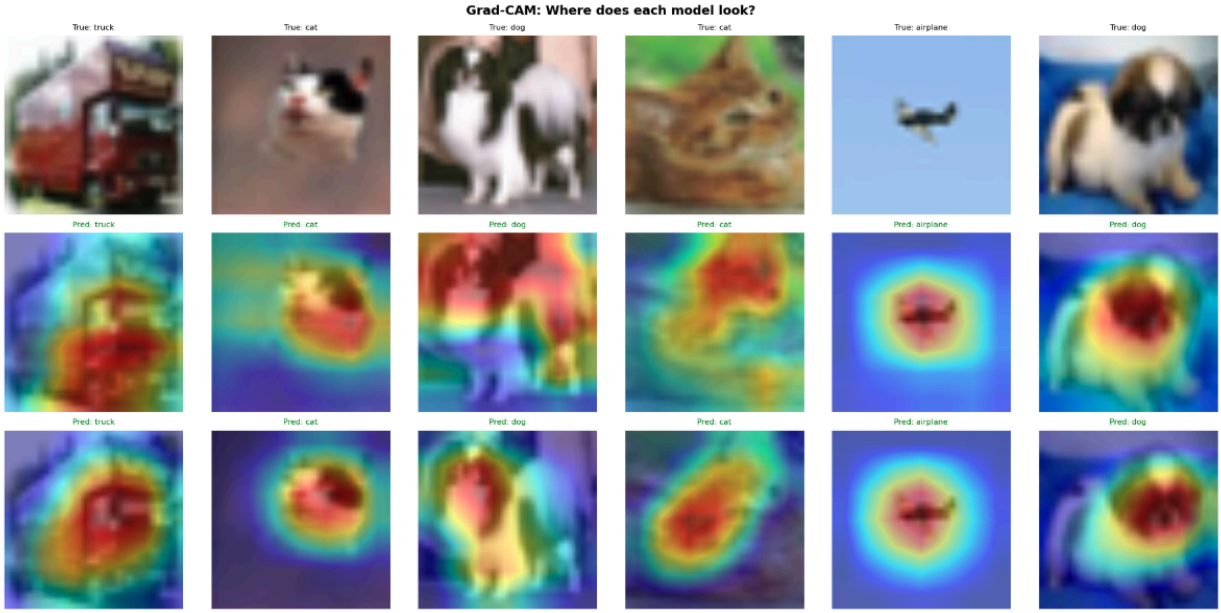


Figure 5. Grad-CAM visualizations on CIFAR-10. Fine-tuning tends to focus more tightly on the object, while Feature Extraction sometimes attends to background regions.

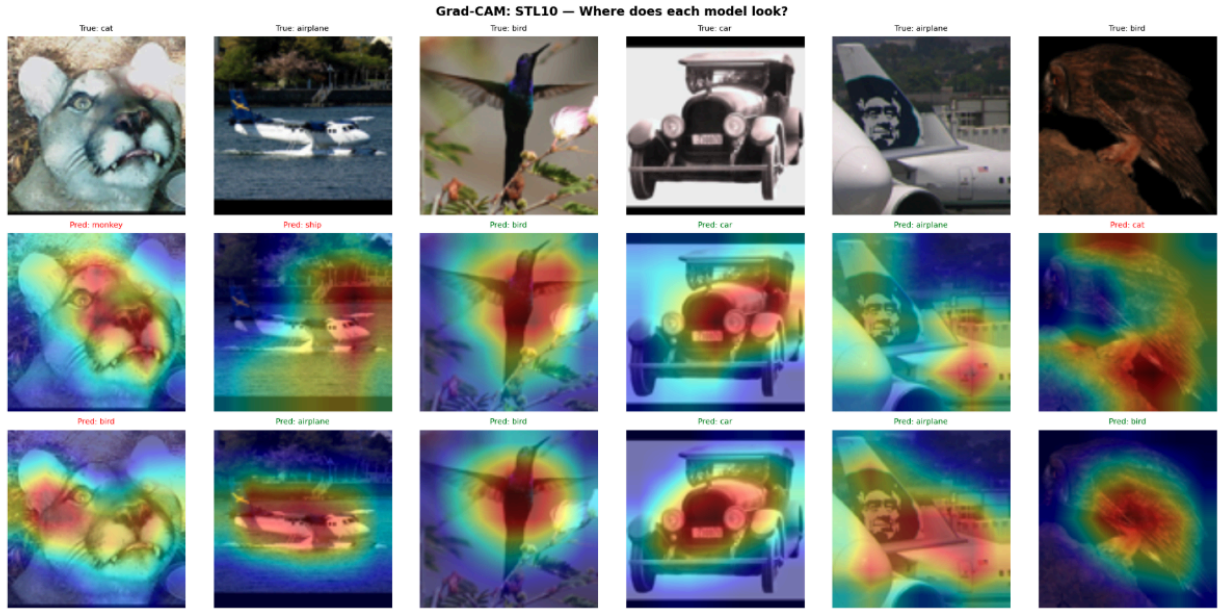


Figure 6. Grad-CAM visualizations on STL10. Both models show focused attention, consistent with their similar accuracy on this dataset.

4. Findings & Analysis

4.1 Effect of Fine-tuning

Fine-tuning consistently performed better than Feature Extraction on both datasets. However, the difference in improvement was striking: +15.76% on CIFAR-10 compared to only +2.37% on STL10.

The main reason for this gap appears to be the significant difference in image resolution. CIFAR-10 consists of very small 32×32 pixel images — about seven times smaller than the 224×224 images used in ImageNet. When these low-resolution images are upscaled, they lose important details and introduce interpolation artifacts. As a result, the low-level features from the pretrained model do not transfer well, and fine-tuning the higher layers (layer 3 and layer 4) becomes much more beneficial.

On the other hand, STL10 images are 96×96 pixels, which is much closer to ImageNet's resolution. Because of this closer match, the pretrained features already worked quite effectively, allowing Feature Extraction to reach a strong 93.16% accuracy. In this case, fine-tuning only provided a small additional boost.

4.2 Feature Extraction: More Data Does Not Always Help

An interesting finding was that Feature Extraction on CIFAR-10 achieved only 79.19% accuracy, even with 45,000 training images. In comparison, the same method on STL10 reached 93.16% using just 4,500 images — ten times fewer.

This clearly shows that simply having more data is not always the key factor. The compatibility between the source domain (ImageNet) and the target dataset, particularly in terms of resolution, seems to play a much bigger role. Better alignment with ImageNet gave STL10 a clear advantage despite the smaller dataset.

4.3 Overfitting in Fine-tuning

During fine-tuning, training accuracy quickly climbed to nearly 99–100% by epochs 5 to 8. However, validation accuracy often plateaued or even dropped slightly toward the end of training. This is a typical sign of overfitting. Saving the model checkpoint with the best validation accuracy — rather than using the final epoch — helped avoid this issue.

4.4 Training Efficiency

Training on STL10 was significantly faster than on CIFAR-10, taking only about 3.2 minutes compared to 25.9 minutes for Feature Extraction. This large difference is mainly due to the much smaller training set (4,500 vs 45,000 images). Interestingly, fine-tuning the additional layers increased training time by only around 6%, suggesting that the extra computational cost is relatively minor compared to the time spent on data loading and forward/backward passes.

5. Conclusions

This project examined two different transfer learning approaches — Feature Extraction and Fine-tuning — with a ResNet18 model pretrained on ImageNet. The evaluation was carried out on CIFAR-10 and STL10, two datasets that share the same 10 classes but differ notably in image resolution and training set size. Overall, the following conclusions can be drawn:

- ➔ Fine-tuning consistently achieves higher accuracy than Feature Extraction, but the benefit is much larger when there is a significant resolution mismatch between the source and target domain (+15.76% for CIFAR-10 vs +2.37% for STL10).

- ➔ Feature Extraction performance depends more on domain similarity than on the quantity of training data. STL10 Feature Extraction (93.16%, 4,500 images) substantially outperforms CIFAR-10 Feature Extraction (79.19%, 45,000 images) due to better resolution alignment with ImageNet.

- ➔ For practical applications, fine-tuning is recommended when the target domain differs in resolution or visual statistics from the pretraining domain. Feature Extraction is sufficient — and computationally cheaper — when the domains are well-aligned.

- ➔ Grad-CAM visualizations confirm that fine-tuning produces more object-focused attention maps, particularly on CIFAR-10, explaining the accuracy improvement in qualitative terms.